DSCI 550: Data Science at Scale

Homework 2, Spring 2026
Due: 11:59 pm, Feb. 18, 2026

1. (30 pts) You have the following training dataset about cars. We want to classify if a car is a family car or not based on the price and engine size.

Car ID	Price	Engine Size	Family Car
1	12000	1000	0
2	23000	2000	1
3	25000	2500	1
4	15000	1500	0
5	56000	3500	0
6	20000	2100	1
7	65000	1000	0
8	30000	1600	1
9	15000	3500	0
10	20000	990	0

   a. (10 pts) Define the most specific hypothesis S as discussed in lecture.
   b. (10 pts) Define the most general hypothesis G as discussed in lecture.
   c. (6 pts) Assuming the most general hypothesis, give an answer if this case is FP, FN, TP, TN.
      a. P= 27000 and S=2000 and Family Car = 1 (Actual)
      b. P= 5000 and S= 3000 and Family = 0 (Actual)
      c. P= 76000 and S=4000 and Family = 1 (Actual)
   d. (4 pts) Now one more data point is added to the training set: < 11, 24500, 2400, 0>. Then, will you change your hypothesis (yes or no)? Then, explain your answer.

2. (20 pts) A binary classifier generates the followings:
   - **Precision = 0.8**
   - **Recall = 0.5**
   - Total actual positive samples = 100

   a) (4 pts) How many **true positives** does the model have?
   b) (4 pts) How many **false negatives** does the model have?
   c) (4 pts) How many **false positives** does the model have?
   d) (4 pts) Compute the **F1 score**.
   e) (4 pts) Explain when and why this kind of high precision and low recall is preferred over low precision and high recall.

   Note: these are theoretical values so real number is possible.

3. (20 pts) (Line separator Question) In the dataset "Grade.xlsx", students had pass/fail grades based on their HW, midterm, and final score. Using this as a training dataset, you are supposed to make a binary classification model to decide if a student with a certain score(s) would pass or fail. Especially, consider only two features (out of three) in your modeling. What would be your model and explain why your model is the best? What is the accuracy of your model with the training dataset? This question is not asking a specific classification algorithm but requiring a conceptual discussion to understand classification. So plain explanation with supporting numbers will be fine as the answer.

4. (30 pts) (Decision Tree) Using the following training dataset, construct a decision tree using Information Gain as discussed in the class. Using V1, V2, V3, V4, predict C.

Data Item	V1	V2	V3	V4	C
1	Yes	Yes	No	Yes	Yes
2	Yes	Yes	Yes	No	Yes
3	No	No	Yes	No	Yes
4	No	Yes	No	Yes	No
5	Yes	No	Yes	Yes	Yes
6	No	Yes	Yes	Yes	No